

## PART A: MULTIPLE CHOICE QUESTIONS

[20 Marks]

Answer the following **TWENTY (20)** multiple choice questions.

1. System sends a fitness alert automatically represent which interaction type?

- ☒ A. Responding
- B. Manipulating
- C. Conversing
- D. Exploring

2. Agile development uses short cycles known as:

- A. Projects
- ☒ B. Sprints
- C. Milestones
- D. Endpoints

3. In user involvement, crowdsourcing refers to:

- A. Hiring more designers
- B. Testing after release only
- C. Asking managers to design
- ☒ D. Getting design ideas from many online contributors

4. Which of the following design techniques help capture user attention?

- A. Increasing screen brightness
- ☒ B. Using colour, sound, spacing
- C. Making more animation
- D. Removing all visual boundaries

5. Which of the following the **BEST** describes a “persona” in requirements gathering?
- A. A visual sketch of a user interface
  - ☒ B. A fictional representation of a target user with goals, behaviours, and characteristics
  - C. A list of functional system requirements
  - D. A quantitative analysis of system performance
6. Which data gathering method involves observing users in their natural environment while performing tasks?
- A. Focus group
  - ☒ B. Contextual inquiry
  - C. Questionnaire
  - D. Studying documentation
7. In task analysis, Hierarchical Task Analysis (HTA) is primarily used to:
- A. Visualize system architecture
  - ☒ B. Break down tasks into sub-tasks to understand workflows
  - C. Conduct usability testing
  - D. Collect demographic information about users
8. Which type of interface is **MOST** suitable for interacting with a smart thermostat via mobile phone?
- A. Command-line interface
  - ☒ B. Graphical user interface
  - C. Tangible interface *→ need physical object*
  - D. Brain-computer interface

9. Which of the following is the main purpose of prototyping in interaction design?

- A. To produce the final version of the system
- ☒ B. To explore and test design ideas before full development
- C. To conduct formal usability testing only
- D. To replace conceptual design entirely

10. Conceptual design primarily focuses on:

- A. The exact programming code of the system
- B. Creating high-fidelity visual mock-ups
- ☒ C. Defining the overall structure, behaviour, and interaction concepts
- D. Implementing SDKs for hardware

11. Concrete design in interaction design is concerned with:

- A. High-level ideas and abstract goals
- B. Evaluating project budgets
- C. Storyboarding future scenarios
- ☒ D. The detailed design of the user interface including screens, layouts, and widgets

12. In interaction design, the scenario is:

- A. A step-by-step instruction manual for coding
- ☒ B. A user-system interaction story
- C. The hardware setup required for the system
- D. A software bug report

13. What is a storyboard used for interaction design?

- A. To show the internal program logic
- ☒ B. To illustrate a sequence of user actions and system responses visually
- C. To test the speed of the final system
- D. To replace the need for usability testing

14. A user journey map is used to:

- ☒ A. Map the user's experience, goals and pain points
- B. Write the final system code
- C. Build a storyboard for hardware prototyping
- D. Determine SDK requirements

15. Which argument supports using Gestalt principles during UI evaluation?

- ☒ A. They explain how users visually group elements
- B. They eliminate the need for usability testing
- C. They ensure system status is always visible
- D. They guarantee error prevention

16. To choose between **TWO (2)** notification styles, which of the following is the most suitable evaluation method?

- ☒ A. A/B testing *→ comparing two version of design to see which perform better*
- ☒ B. Gestalt evaluation *→ evaluation principle only*
- C. Predictive modelling
- D. Walkthrough

17. Which of the following shows the difference between heuristic evaluation and walkthroughs?

- ☒ A. Only heuristic evaluations <sup>don't always</sup> require expert evaluators
- ☒ B. Walkthroughs follow pre-planned scenarios, heuristics do not
- C. Walkthroughs rarely identify usability problems
- D. Heuristic evaluations must use users

18. Why does Nielsen recommend **FIVE (5)** evaluators for heuristic evaluation?

- A. Because it meets ISO standards
- ☒ B. Because ~75–80% of problems can be found with five evaluators
- C. Because more than five reduces reliability
- D. Because fewer than five always misses critical errors



19. During an evaluation of a medical prescription app, you observe that nurses frequently choose the wrong medication dosage because **TWO (2)** dosage options appear very similar in layout and formatting. As an evaluator, which heuristic is most relevant for identifying this problem?

- A. Aesthetic and minimalist design
- B. Flexibility and efficiency
- ☒ C. Error prevention
- D. Recognition rather than recall

20. Which of the following contributes to the severity of a usability problem?

- I. Frequency of the problem
- II. Cause of the problem
- III. Impact of the problem
- IV. Persistence of the problem

- A. I and II
- B. II and III
- ☒ C. I, II and III
- ☐ D. I, III and IV

## PART B: STRUCTURED QUESTIONS

[33 Marks]

This part consists of **FOUR** short and structured questions. The marks for each part of the question are as indicated.

### QUESTION 1

(11 marks)

You are asked to take the responsibility of designing a machine. This machine can be worked as a printer and a scanner. It can also be used as a fax and a copy machine. It can be installed at home, or in small offices.

- a) To ensure the designing of the product fulfils the usability criteria, you decide to include the four basic activities in Interaction Design in your project plan. What are these **FOUR** basic activities? List all of them. (4 marks)
- b) One of the practical issues in a project is to identify who are the users/stakeholders of a product. In the case of your project stated above, who would be your users/stakeholders? List **FOUR** of them. (4 marks)
- c) **List and briefly** explain **THREE** quality thresholds that will help you choose among the design alternatives for your product. (3 marks)

### QUESTION 2

(6 marks)

Low fidelity prototyping (LFP) uses medium which is unlike the final medium. It is often used with scenarios, bringing more detail, and a chance to role play. One of the approaches used for LFP is the *Wizard of Oz*.

- a) Describe what is the *Wizard of Oz* (4 marks)
- b) Give an example to how and when this approach is best used. (2 marks)

a) establishing requirements  
designing alternative  
prototyping  
evaluating

b) 1) home users  
2) office employees  
3) IT administrators  
4) Students

c) 1) Usability - product should be easy to learn, use and efficient for users to complete their task

2) UX - product should provide a satisfying, enjoyable experience while interacting with it.

3) Utility - whether the product provides the right functionality to help user achieve specific task.

## Question 2

a) ITP technique where users believe they are interacting with fully functioning system, but the system is actually operated manually by a person hidden from the user.

b) example: designing a voice control smart home assistant  
- instead of an AI understanding a user command, a hidden researcher listen and manually activate ... using appropriate ...  
- during early stages of system development.

## Question 3

A.3a → Similarity

A.3b → Enclosure

A.3c → Figure and ground

A.3d → Continuity

## Question 4

a) Heuristic Evaluation

usability testing

Conduct by usability expert

conduct by real user

Expert conduct the interface using Nielsen's 10 heuristics.

users perform real task while observer record their performance

Does not require user

require representative user

identify usability problem based on design principle

identify usability problem based on actual user behavior and feedback

b) 1) visibility of system status

- the system should always keep users informed about what is happening by clear and timely feedback

ex: After a user uploads a file, the system displays:  
uploading ... 75% complete:

2) error prevention

- the interface should be designed to prevent mistakes before they happened instead of displaying error message afterward.

ex: before deleting an important document the system ask:  
are you sure you want to delete this file?  
with yes and cancel button

### QUESTION 3

(4 marks)

Gestalt principles or laws are rules that describe how the human eye perceives visual elements. These principles aim to show how complex scenes can be reduced to more simple shapes. They also aim to explain how the eyes perceive the shapes as a single, united form rather than the separate simpler elements involved. Given the following figures A.3a, A.3b, A.3c and A.3d, name the correct Gestalt principle used, respectively.

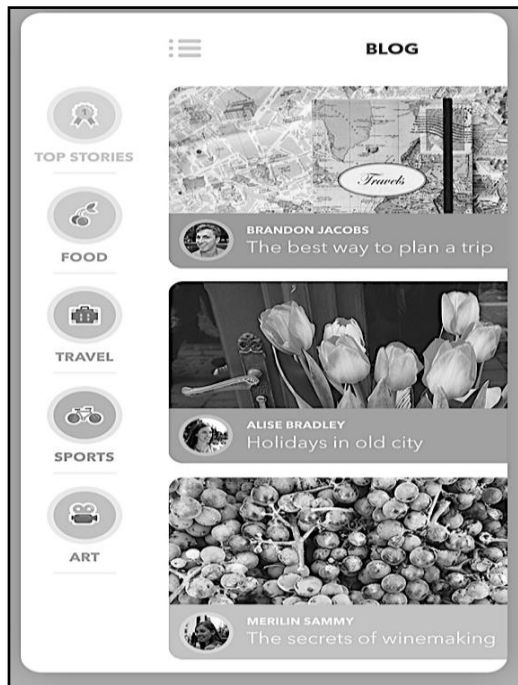


Figure A.3a.



Figure A.3b.

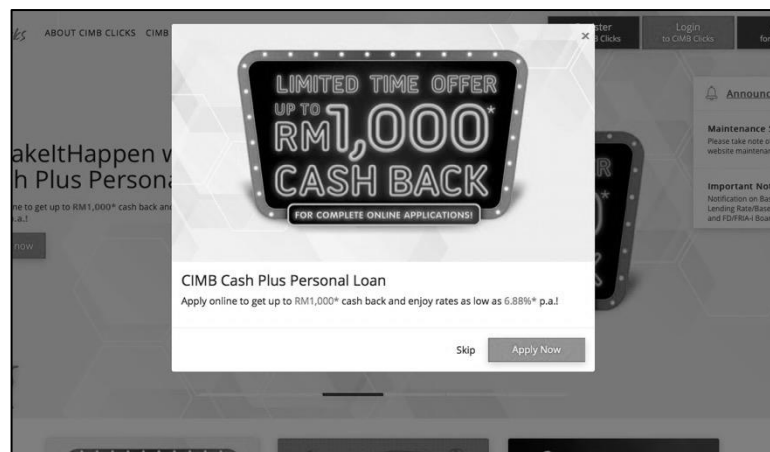


Figure A.3c.

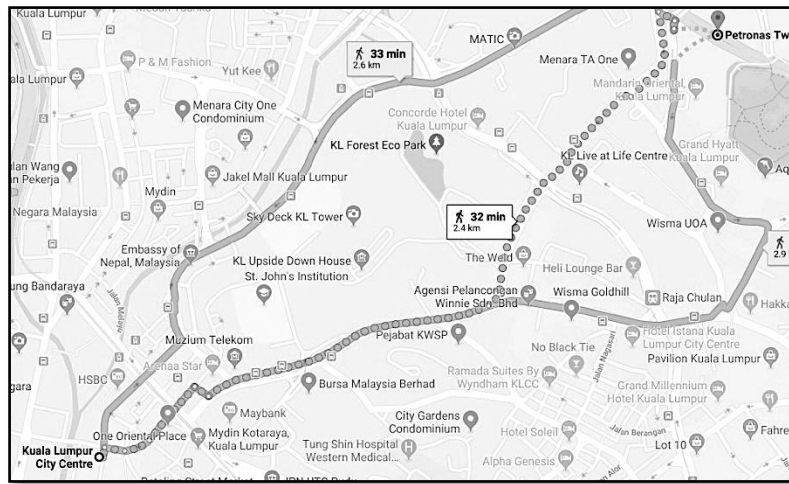


Figure A.3d.

#### QUESTION 4

(12 marks)

Heuristic evaluation and usability testing are two different techniques for finding usability problems.

- Explain the difference between Heuristic Evaluation and Usability Testing. (4 marks)
- List and describe **TWO** of Nielsen's heuristic evaluation (design guideline principle) by providing **ONE** example to support each answer. (8 marks)

**[67 Marks]**

## QUESTION 1

**(34 marks)**

Based on IoT Agenda<sup>1</sup>, the Internet of Things (IoT) is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction. See Figure B.1.

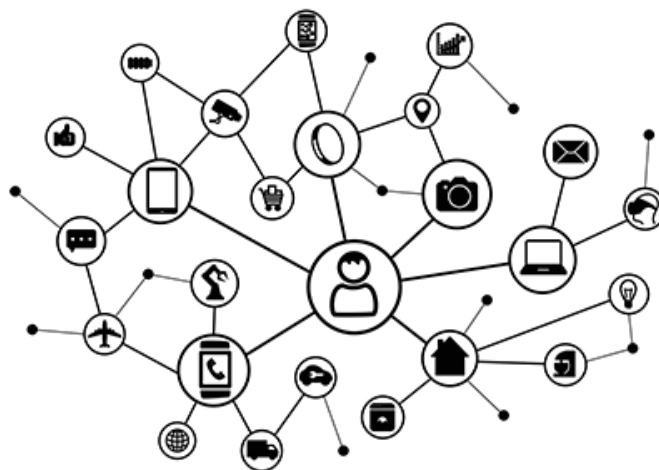


Figure B.1. Internet of Things (credit Champion Solutions Group)

The company understands that the rush decision to develop any IoT project is not wise. Hence, your team is responsible to perform an insight study to understand about the users, their behaviours and their needs, before arriving to recommendations. Your first IoT project will be implemented in the context of higher institutions.

Based on the above, answer the following questions.

<sup>1</sup> “What is Internet of Things (IoT)? – Definition from WhatIs.com.” <http://internetofthingsagenda.techtarget.com/definition/Internet-of-Things-IoT>. Accessed 26 Nov. 2019.

- a) What is the importance of understanding your users before embarking with IoT implementation? (2 marks)
- b) You only have a week to perform the insight study. What TWO techniques would you find to be the most suitable to gather information about your users? Why? Elaborate by including the recorded data type and equipment used. (7 marks)
- c) How do you represent your users from the data gathered? What are the most important elements or characteristics to capture? (3 marks)
- d) Based on the context of IoT at higher institution, distinguish TWO targeted users you found to be most important to be considered in (d), and suggest TWO goals. (5 marks)
- e) From your answers in (d), provide TWO scenarios by sketching via storyboarding, how each user achieving their goals. Storyboard should convey setting (people, environment, task), sequence (steps, shows interactions), and satisfaction. (12 marks)
- f) For each scenario in (e), justify the reasons why you pursue for the chosen interaction and interface types. You may choose 1 interface and 1 interaction type for each scenario. (5 marks)

## **QUESTION 2**

**(33 marks)**

Figure B.2 shows the front page of the Universiti Teknologi Malaysia' Counselling Center website. You have been asked to re-design or design a new version of the page that can be viewed correctly on multiple screens including desktop and mobile phone.

- a) How does the page measure up against UI/UX design principles? Which part of it is good? Which part of it is bad? Apply any **THREE** design principles/heuristics that you have learned to support your answer. (6 marks)
- b) You have decided to setup a survey asking prospective student, current student, staff, alumni and visitor on how they use the website to identify the most commonly used elements. How would you encourage good questionnaire response? List **FOUR** ways to do so. (4 marks)
- c) Describe **THREE** specific problems or research designs issues that need to be considered when designing for mobile phone screen. (6 marks)
- d) How would you evaluate your new design? Give a **thorough** description on how you would execute a usability study for empirical evaluation. Your answer should also include specification on what to measure including independent and dependent variables, and under what conditions, and how to avoid order effects. (17 marks)



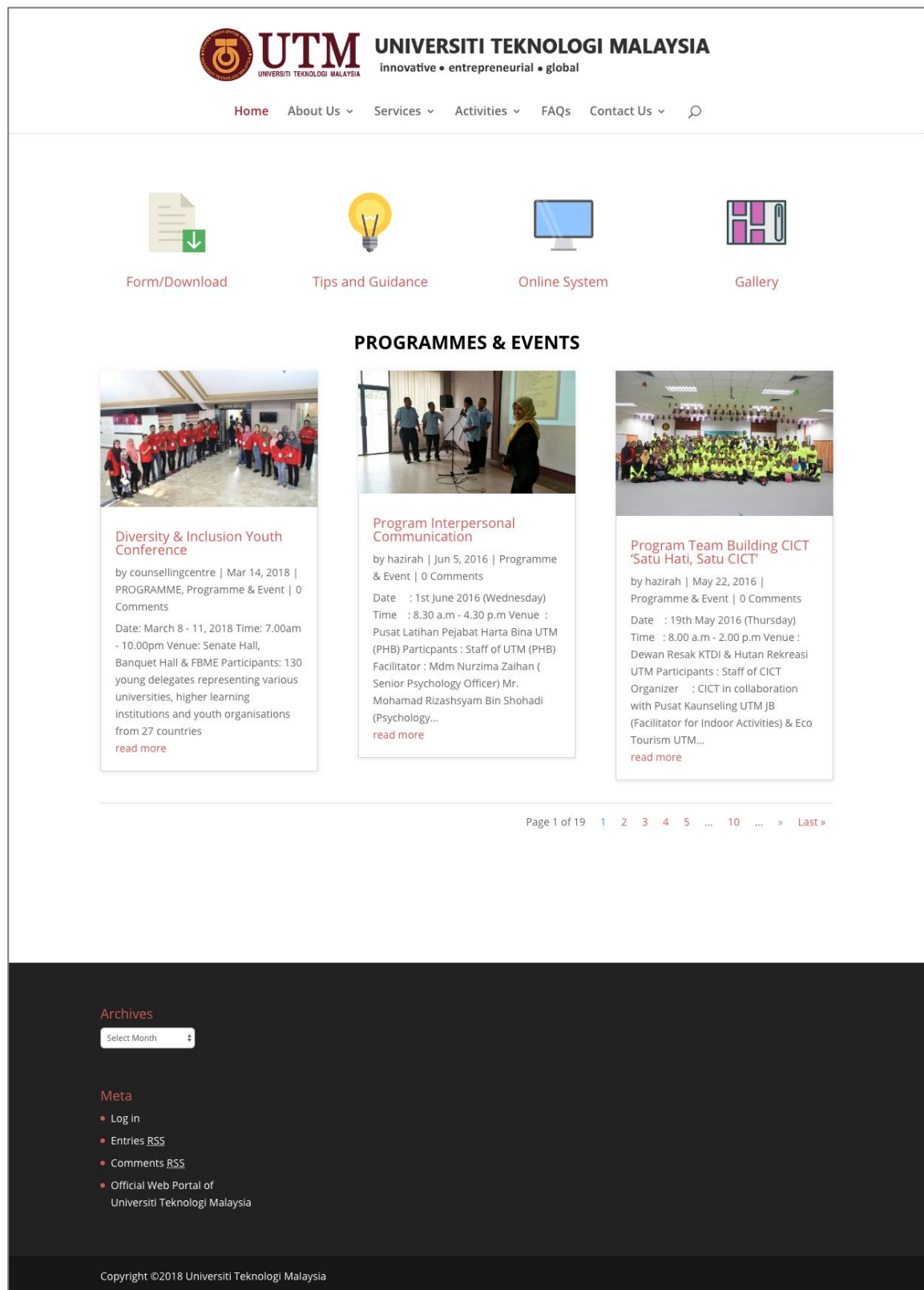


Figure B.2: The front page of Universiti Teknologi Malaysia's Counselling Center website

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